



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2421151

Name of the Program : BCA/BCA AI/DS / B.Sc. IT

Semester: 04

Name of the Course : Web Application Development

Course Code: TBC /TBD 401 / TBI 401

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

- a. Explain the concept of hyperlinks in XHTML. Write an XHTML program that demonstrates the use of the anchor (<a>) tag, including attributes such as href, target, and title, with suitable examples. CO1
- b. Explain the role of web browsers in accessing internet resources. Discuss the different types of web browsers and describe how a browser communicates with a web server to retrieve and display a web page. CO1
- c. Write an XHTML program to demonstrate different types of lists. Include ordered list, unordered list, and definition list with suitable examples. CO1

Q2.

(2X10=20 Marks)

- a. Explain the different types of CSS, Inline CSS, Internal CSS, and External CSS. Write suitable examples to demonstrate each type. CO2
- b. Design an XHTML form for user feedback or login/registration. Also explain the purpose of each form element used. CO2
- c. Explain the concept of the CSS Box Model. Describe its components, namely content, padding, border, and margin, with a suitable diagram and example. CO2

Q3.

(2X10=20 Marks)

- a. Design an XHTML form and write JavaScript code to perform form validation. Validate the following:
 - Name (should not be empty)
 - Email (valid format)
 - Password (minimum 6 characters)
 - Mobile number (only digits, 10 characters)

Display appropriate error messages and prevent form submission if validation fails.

CO3

- b. Explain the concept of events and events registration in JavaScript. Write a program to demonstrate different types of events such as:
 - onclick
 - onmouseover
 - onchange

CO3

- c. Write a JavaScript program to create an array named course using different methods and initialize it with the elements "BCA", "MCA", "BSCIT", and "MSCIT". Perform the following operations: CO4

1. Access and display array elements
2. Modify an existing element (e.g., change "MCA" to "MBA")



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3. Insert a new element (e.g., "B.Tech")
4. Delete an element from the array
5. Traverse the array using loops (for, forEach)

Q4.

(2X10=20 Marks)

- a. Explain different types of arrays in PHP (indexed, associative, and multidimensional). Write a program demonstrating operations on each type. CO5
- b. Write a PHP program to perform CRUD operations (Create, Read, Update, Delete) using a MySQL database. Create a sample table (e.g., student with fields like id, name, course) and demonstrate all four operations with proper database connectivity. CO5
- c. Explain the concept of string handling in PHP. Write a PHP program demonstrating at least four string functions such as strlen(), strrev(), strtoupper(), strpos(), etc. CO4

Q5.

(2X10=20 Marks)

- a. Discuss the advantages of using the MERN Stack for web development. Explain why it is widely adopted in modern web applications, with relevant examples. CO6
- b. Explain how different components of the MERN stack work together in a typical application. Describe the flow of data from the user interface to the database and back. CO6
- c. Define Full Stack Web Development. Explain the roles of the front-end, back-end, and database layers in building a web application, with suitable real-world examples. CO5

Note For the question paper setters:

- Question paper should cover all the COs of the course, preferably equal distribution.
- Please specify COs against each question.